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Week 3

Ant colony optimization for the travelling salesman problem.

Warehouse robot navigation.

Application: Robotics

problem Statement: Determine the collision-free paths for robots moving between storage shelves and packing stations while reducing travel time and avoiding congested aisles.

Objective: Minimize travel time and path length

Algorithm:

Step 1: Create a grid map of the warehouse floor containing a start shelf, a packing station, solid obstacles, and high-traffic congested aisle.

Step 2: Initialize the parameters like the number of virtual ants, path visibility and initial pheromone values on the floor grid.

Step 3: place all virtual ants at the start shelf to begin their pathfinding simulation toward the destination.

Step 4: Move each ant step by step to a neighboring cell by calculating a probability based on high pheromone levels and low traffic costs.

Step 5: complete the ant's movement once it either successfully reaches the packing station or gets trapped in dead-end.

Step 6: Evaporate a small percentage of pheromones globally accross the entire grid to deleted old, slow and inefficient routes.

Step 7: Deposit fresh, heavy pheromone reward only on the paths used by successful ants, giving highest reward to the fastest, traffic free route.

Step 8: Repeat this entire process for a fixed number of rounds until all virtual ants converge onto the exact same winning path.

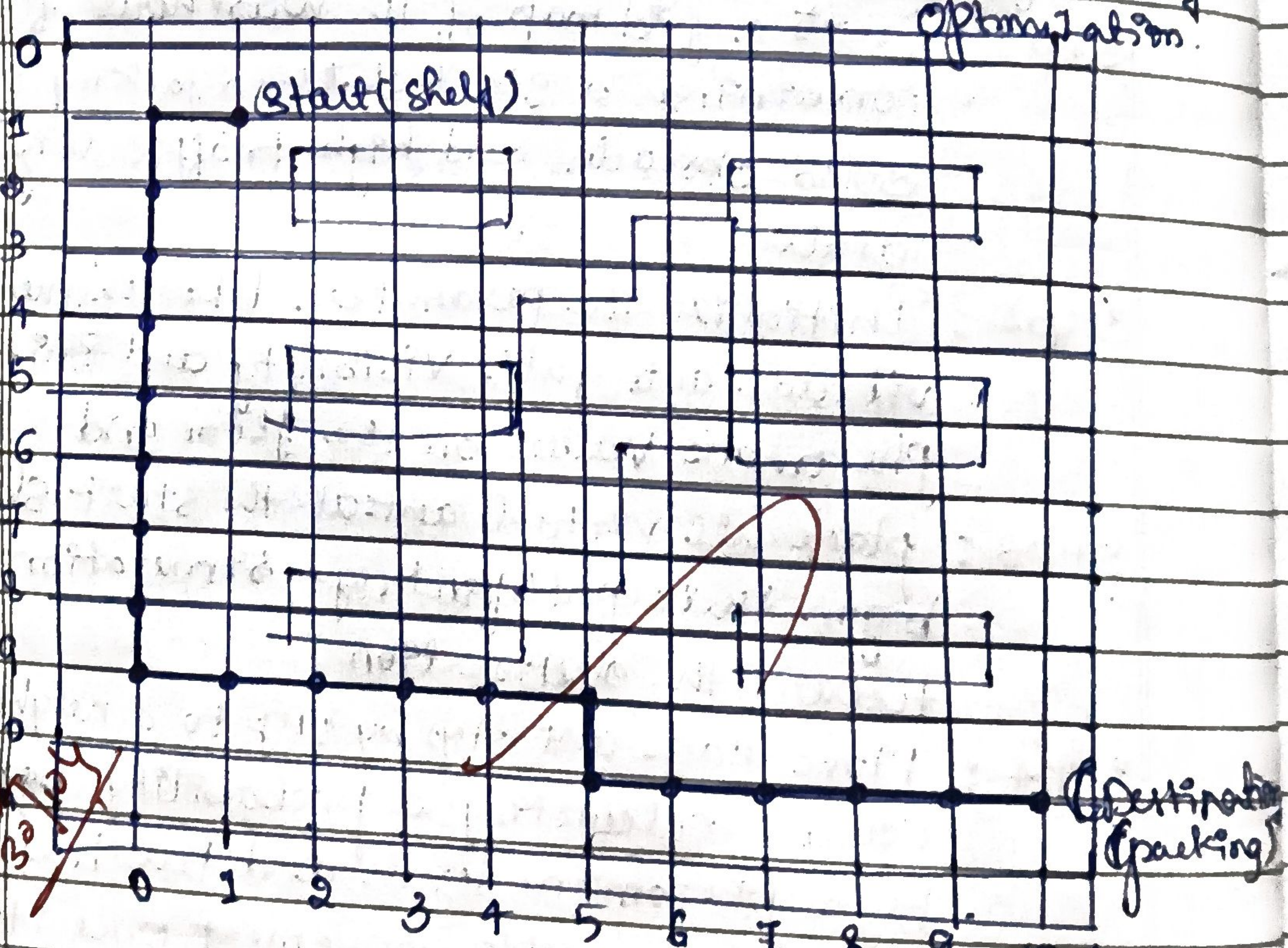
Step 9: Output and plot the final optimal, collision-free and lowest travel-time route for the actual warehouse robot to execute.

Procedure & 30/10/24

Output:

Optimization complete | Best path travel cost/time: 20.00

Warehouse Robot Navigation via Ant colony Optimization



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Week 4

Cuckoo Search Algorithm

Data center cooling optimization

Application: Cloud infrastructure and energy efficiency

Problem statement: A data center has multiple cooling units with adjustable airflow rates and temperature settings. Develop CS algorithm to determine suitable cooling configurations based on server temperatures, heat loads and cooling energy consumption.

Objective : Minimize cooling energy consumption while maintaining safe server operating temperatures

Algorithm:

Step 1: Define the problem by representing a 'Nest' as a cooling configuration.

Step 2: Define the fitness function to calculate total cooling consumption adding a massive penalty if any server exceeds its safe operating temperature.

Step 3: Initialize parameters including the number of nests, the discovery probability (Pa = 0.25) and maximum iteration

Step 4: Generate an initial population of nests with random airflow rates and temperature settings within safe ~~operational~~ operational boundaries.

Step 5: Evaluate the fitness of every nest to find the current best cooling configuration that saves energy. [illegible] while keeping server cool.

Step 6: Generate new cooling configurations by performing Levy flights to make smart exploratory tweaks to the airflow and temperature values.

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Step 7 : Compare the new configuration against the old ones and replace an old nest if the new cuckoo egg has better energy efficie

Step 8 : Abandon a fraction (Pa) of the worst-pe high energy nests and replace them with completely fresh ~~freedom~~ random co Setups.

Pro ced Step 9 : Repeat the evaluation, Levy flights and replacement steps for fixed number of iterations until cooling system configura-tion stabilizes.

30/10

Step 10 : Output and plot the optimal airflow rates and temperature settings that minimize the power consumption while guaranteeing safe server temperatures

o/p

Output:

optimization complete:
Minimized cooling power : 4411.01 kW
Final Server Temperature : 25.41°C
Cooling Unit 1 -> Airflow : 568.6 m³/h | Temp : 1
Cooling unit 2 -> Airflow : 592.7 m³/h | Temp : 20
Cooling unit 3 -> Airflow : 919.2 m³/h | Temp : 17.4

Data Center Energy optimization